

new_alt

June 25, 2025

```
[1]: import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.linear_model import LogisticRegression
import matplotlib.pyplot as plt
from scipy import stats
import plotly.graph_objects as go
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.manifold import TSNE
import seaborn as sns
import shap
import plotly.express as px
from sklearn.metrics import (accuracy_score, roc_auc_score,
    ↪classification_report,
                                confusion_matrix, roc_curve, f1_score)

from sklearn.ensemble import RandomForestClassifier,
    ↪HistGradientBoostingClassifier
from sklearn.inspection import permutation_importance
from sklearn.cluster import KMeans
import sys
sys.path.append('src')
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/site-
packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update
jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user_install.html
    from .autonotebook import tqdm as notebook_tqdm
```

```
[2]: df = pd.read_csv('myopia.csv', sep=';')
df_train = pd.read_csv('myopia_train.csv', sep=',')
df_test = pd.read_csv('myopia_test.csv', sep=',')
df
```

```
[2]:
```

	ID	STUDYYEAR	MYOPIC	AGE	GENDER	SPHEQ	AL	ACD	LT	VCD	\
0	1	1992	1	6	1	-0.052	21.89	3.690	3.498	14.70	
1	2	1995	0	6	1	0.608	22.38	3.702	3.392	15.29	

2	3	1991	0	6	1	1.179	22.49	3.462	3.514	15.52
3	4	1990	1	6	1	0.525	22.20	3.862	3.612	14.73
4	5	1995	0	5	0	0.697	23.29	3.676	3.454	16.16
..	...	...	...	...	...	...	...	...	...	...
613	614	1995	1	6	0	0.678	22.40	3.663	3.803	14.93
614	615	1993	0	6	1	0.665	22.50	3.570	3.378	15.56
615	616	1995	0	6	0	1.834	22.94	3.624	3.424	15.89
616	617	1991	0	6	1	0.665	21.92	3.688	3.598	14.64
617	618	1994	0	6	0	0.802	22.26	3.530	3.484	15.25

	SPORTH	READHR	COMPHR	STUDYHR	TVHR	DIOPTERHR	MOMMY	DADMY
0	45	8	0	0	10	34	1	1
1	4	0	1	1	7	12	1	1
2	14	0	2	0	10	14	0	0
3	18	11	0	0	4	37	0	1
4	14	0	0	0	4	4	1	0
..	...	...	...	...	...	...	...	...
613	2	0	7	3	14	37	1	0
614	6	0	1	0	8	10	1	1
615	8	0	0	0	4	4	1	1
616	12	2	1	0	15	23	0	0
617	25	0	2	0	10	14	1	1

[618 rows x 18 columns]

```
[11]: cat = ['MYOPIC', 'GENDER', 'PARENTSMY']
myopianum = df_train.drop(cat, axis=1)
myopiafact = df_train[cat]

myopianum.head(5)

myopiafact.head(5)
```

```
[11]: MYOPIC  GENDER  PARENTSMY
0         0         0           1
1         0         0           1
2         0         0           1
3         1         1           1
4         0         0           1
```

0.0.1 Univariate Analysis and Multivariate Visualization

- **Distribution of Continuous Features:** Visualized using boxplots and histograms.
- **Categorical Features Breakdown:** Analyzed to understand their distribution and impact.

```
[12]: myopianum.hist(figsize=(15,10))
plt.tight_layout()
plt.show()
```

```

myopiafact.hist(figsize=(15,10))
plt.tight_layout()
plt.show()

num_cols = df_train.drop(cat, axis=1).columns.tolist()
for col in num_cols:
    df_train[col] = pd.to_numeric(df_train[col], errors='coerce')

df_train['GENDER'] = df_train['GENDER'].astype('category')
df_train['PARENTSMY'] = df_train['PARENTSMY'].astype('category')
df_train['MYOPIC'] = df_train['MYOPIC'].astype('category')

# crosstab "MYOPIC" x "GENDER"
print("MYOPIC x GENDER")
print('Nb Boys : ', (df_train['GENDER']==0).sum())
print('Nb Girls : ', (df_train['GENDER']==1).sum())
# Chi-2 test on this crosstab
table1 = pd.crosstab(df_train['MYOPIC'], df_train['GENDER'])
chi2, p, dof, ex = stats.chi2_contingency(table1)
print('p-value chi-2:', p)
print('chi-value chi-2:', chi2)
print('dof-value chi-2:', dof)
print('ex-value chi-2:', ex)
# crosstab
table_genre = pd.crosstab(df_train['MYOPIC'], df_train['GENDER'],
    ↪normalize='columns').round(2)
table_genre.index = ['No-Myopic', 'Myopic']
table_genre.columns = ['Man', 'Woman']
display(table_genre)

# crosstab "MYOPIC" x "PARENTMY"
print("\n", "-" * 30, "\nMYOPIC x PARENTSMY")
print('Nb Myopic Parents : ', (df_train['PARENTSMY']==1).sum())
print('Nb Non Myopic Parents : ', (df_train['PARENTSMY']==0).sum())
# Chi-2 test on this crosstab
table2 = pd.crosstab(df_train['MYOPIC'], df_train['PARENTSMY'])
chi2, p, dof, ex = stats.chi2_contingency(table2)
print('p-value chi-2:', p)
print('chi-value chi-2:', chi2)
print('dof-value chi-2:', dof)
print('ex-value chi-2:', ex)

# Crosstab
table_Parents = pd.crosstab(df_train['MYOPIC'], df_train['PARENTSMY'],
    ↪normalize='index').round(2)
table_Parents.index = ['No-Myopic', 'Myopic']

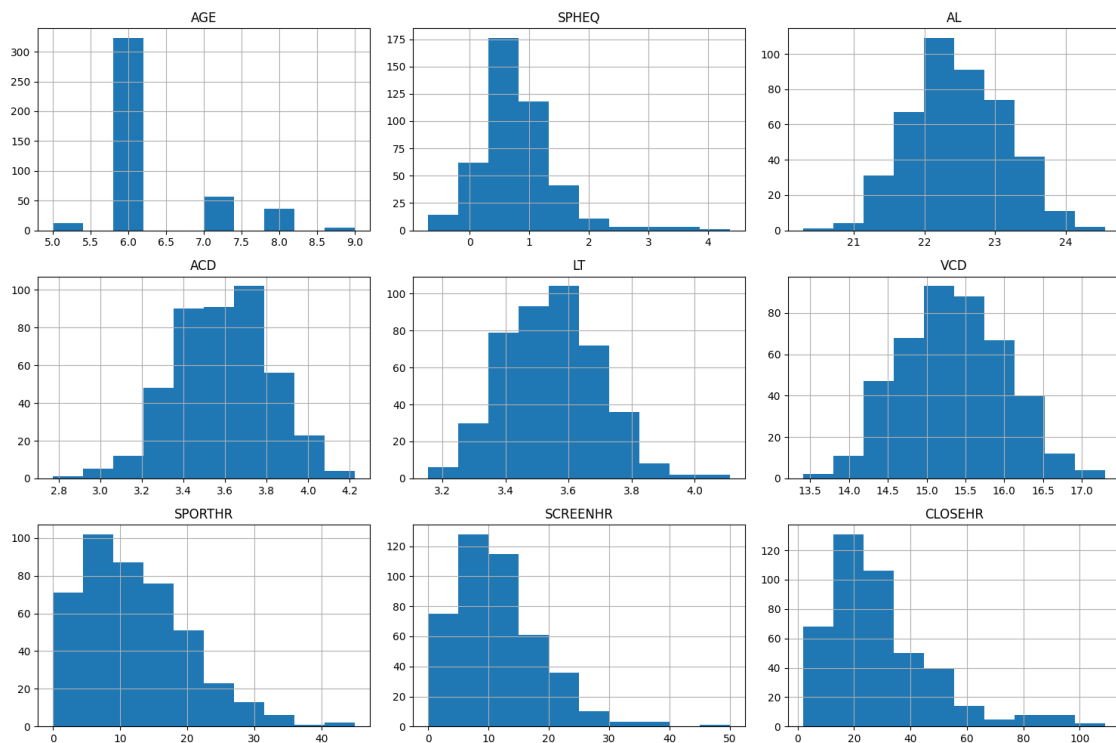
```

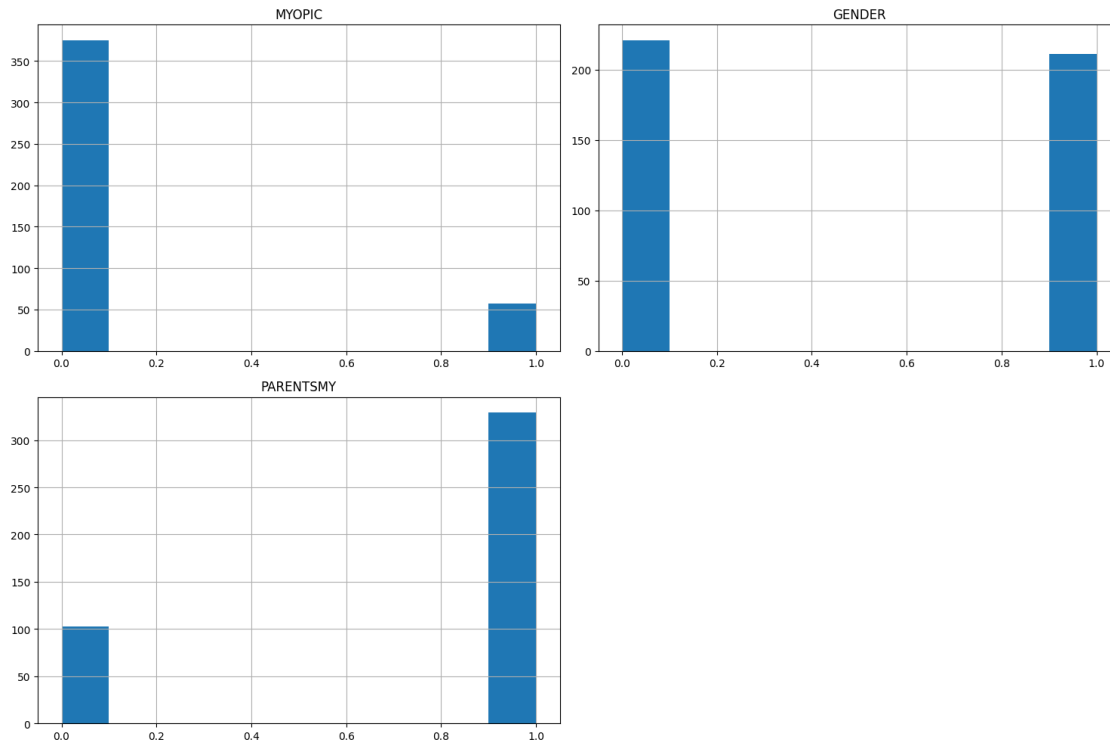
```

table_Parents.columns = ['No', 'Yes']
display(table_Parents)

fig = go.Figure(data=[
    go.Bar(name='GENDER',
           x=['Males', 'Females'],
           y=table_genre.loc['Myopic'].values*100),
    go.Bar(name='PARENTMY',
           x=['No Myopic Parents', 'At least one Myopic Parent'],
           y=table_Parents.loc['Myopic'].values*100)
])
fig.update_layout(barmode='group', yaxis_title="Percentage of Myopic(%)")
fig.show()

```





MYOPIC x GENDER

Nb Boys : 221

Nb Girls : 211

p-value chi-2: 0.29795054292322165

chi-value chi-2: 1.0833446284321815

dof-value chi-2: 1

ex-value chi-2: [[191.84027778 183.15972222]
[29.15972222 27.84027778]]

	Man	Woman
No-Myopic	0.89	0.85
Myopic	0.11	0.15

MYOPIC x PARENTSMY

Nb Myopic Parents : 329

Nb Non Myopic Parents : 103

p-value chi-2: 0.00021562337360903677

chi-value chi-2: 13.689791775451848

dof-value chi-2: 1

ex-value chi-2: [[89.40972222 285.59027778]
[13.59027778 43.40972222]]

	No	Yes
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No-Myopic	0.27	0.73
Myopic	0.04	0.96

crosstab	p value	chi2	Proportion Analysis
Myopic-Gender	0.3	1.1	p-value above 0.05. It implies that there is no statistically significant association between gender and myopia.
Myopic-Parentsmy	2.1 E-5	13.2	A highly significant p-value shows a strong association between parental myopia and child myopia. Children with at least one myopic parent are much more likely to be myopic themselves.

Analysis of Gender and Parental Myopia Association with Myopia Status

Gender and Myopia:

No statistically significant association was found between gender and myopia incidence ($p > 0.05$).

Boys and girls have similar rates of myopia in this population.

Parental Myopia:

A strong and highly significant association was found between having at least one myopic parent and being myopic ($p < 0.001$).

Children with myopic parents are much more likely to develop myopia themselves.

Implications:

Gender does not appear to be a risk factor for myopia in this dataset.

Parental (hereditary) myopia should be prioritized when assessing a child's risk for developing myopia.

```
[13]: ### 2.1.2. Quant values

for col in myopianum.columns:
    group0 = df_train.loc[df_train['MYOPIC'] == 0, col]
    group1 = df_train.loc[df_train['MYOPIC'] == 1, col]
    stat, p = stats.ttest_ind(group0, group1, nan_policy='omit')
    fig = px.box(df_train, x='MYOPIC', y=col, color='MYOPIC', points="all",
    ↪title=col)
    fig.show()
```

```
print(f"T-test {col}: statistic={stat:.2f}, p-value={p:.4f} \n-----")
```

T-test AGE: statistic=0.41, p-value=0.6815

T-test SPHEQ: statistic=8.91, p-value=0.0000

T-test AL: statistic=-0.46, p-value=0.6479

T-test ACD: statistic=-2.67, p-value=0.0078

T-test LT: statistic=1.50, p-value=0.1345

T-test VCD: statistic=0.11, p-value=0.9125

T-test SPORTHR: statistic=2.13, p-value=0.0335

T-test SCREENHR: statistic=0.40, p-value=0.6871

T-test CLOSEHR: statistic=0.19, p-value=0.8506

Col	NonMyopic BoxPlot	Myopic Boxplot	Test-T – p-value	Test-T – statistic	Conclusion
SPORTHR	q1 6, median 10, q3 16	q1 3, median 8, q3 15	p-value = 0.0335	2.13	Myopia ⇒ less sport
ACD	q1 3.42, median 3.57, q3 3.53	q1 3.52, median 3.69, q3 3.8	p-value = 0.0078	-2.67	Myopia ⇒ less sport
SPHEQ	q1 0.545, median 0.791, q3 1.097	q1 -0.0735, median 0.234, q3 0.507	p-value = 0.0000	8.91	Myopia ⇒ lower SPHEQ

Statistical Comparison of Myopic and Non-Myopic Groups

T-Test analysis The t-test evaluates whether two groups have significantly different means. A **negative t-statistic** indicates that the mean of the first group is lower than the second. The **p-value** quantifies the likelihood that this difference is due to chance. A **small p-value (typically < 0.05)** suggests a statistically significant difference between groups, regardless of the sign of the t-statistic.

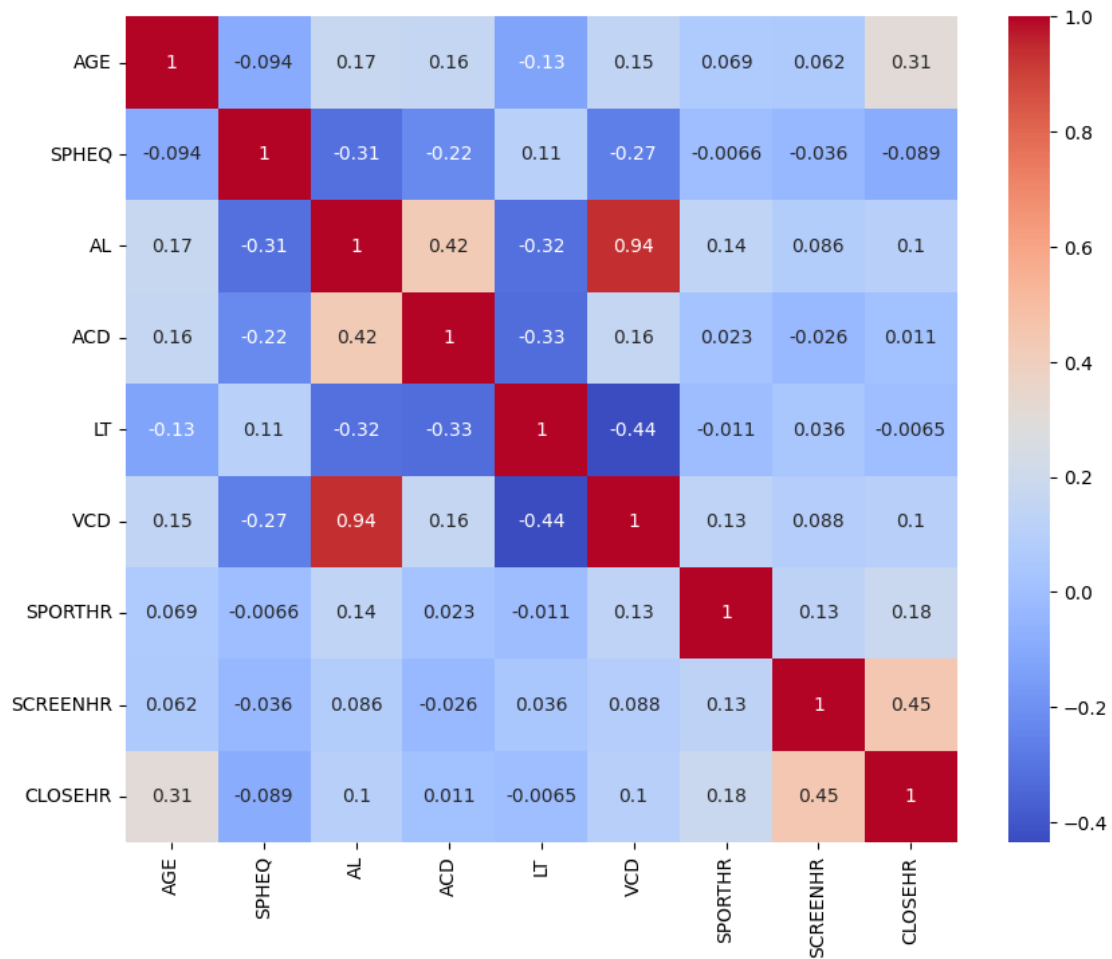
```
[16]: corr = myopianum.corr()
plt.figure(figsize=(10,8))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.show()

# Suppression des doublons et diagonales
corr_pairs = (
    corr.where(np.triu(np.ones(corr.shape), k=1).astype(bool))
        .stack()
        .reset_index()
)
corr_pairs.columns = ['Variable 1', 'Variable 2', 'Correlation']

# Tri du plus faible au plus fort
corr_pairs_sorted = corr_pairs.reindex(corr_pairs['Correlation'].abs().
    ↪sort_values(ascending=False).index)

high_correlation = corr_pairs_sorted[corr_pairs_sorted['Correlation'].abs() > 0.
    ↪4 ]
display(high_correlation)
display(("Sorted correlations between variables (excluding self-correlations)"))

# Création d'une liste unique des variables fortement corrélées
var = high_correlation['Variable 1'].unique().tolist() + [
    val for val in high_correlation['Variable 2'].unique()
    if val not in high_correlation['Variable 1'].unique()
]
sns.pairplot(df_train, hue='MYOPIC', vars=var)
plt.suptitle("Pairplot key variables", y=1.02)
plt.show()
```



	Variable 1	Variable 2	Correlation
17	AL	VCD	0.942082
35	SCREENHR	CLOSEHR	0.449825
26	LT	VCD	-0.435219
15	AL	ACD	0.423216

'Sorted correlations between variables (excluding self-correlations'



[]:

[]: